

Stationary probability kernels need not be disintegrations

A finite counterexample to Remark 3.13

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Result. The answer to Amrutam–Klötzer–Oppelmayer’s Remark 3.13 is negative. Stationarity and the barycenter identity do not force a measurable probability kernel to be the disintegration of a factor map, even for finite spaces and trivial actions.

1 Source question

For a separable Γ - C^* -algebra $\mathcal{A}$ and a measurable Γ -space (Y, η) , Proposition 3.12 of [1] produces a μ -stationary measurable field of states. Taking $\mathcal{A} = C(X)$ gives a measurable map

$$\varphi : Y \longrightarrow \text{Prob}(X),$$

and its barycenter is

$$\nu = \int_Y \varphi(y) d\eta(y).$$

Remark 3.13 asks whether φ is the disintegration map associated with some factor map $\pi : (X, \nu) \rightarrow (Y, \eta)$.

2 Idea of the counterexample

A disintegration has a support condition absent from stationarity: if $\{\nu_y\}$ disintegrates ν along π , then

$$\nu_y(\pi^{-1}(y)) = 1 \quad \text{for } \eta\text{-almost every } y.$$

With trivial group actions every probability kernel is stationary. We may therefore choose a kernel whose measures overlap completely across distinct values of y , making fiber support impossible. A singleton domain already rules out the existence of any factor; a two-point variant shows that the failure persists even when factor maps do exist.

3 Counterexamples

Theorem 1. *There are finite compact Γ -spaces X, Y , an admissible non-degenerate probability measure μ on Γ , and a μ -stationary measurable map $\varphi : Y \rightarrow \text{Prob}(X)$ such that, for $\nu = \int_Y \varphi(y) d\eta(y)$, no factor map $\pi : (X, \nu) \rightarrow (Y, \eta)$ has φ as its disintegration. One may choose the example so that no factor exists at all, or so that factors exist but the prescribed field is not their disintegration.*

Given a probability measure $\mu \in \text{Prob}(\Gamma)$, a fixed point in $\mathcal{C}_Y$ under the convolution of μ is called μ -stationary. That such a stationary map always exists is a standard Markov-Kakutani fixed point argument.

Proposition 3.12. *Let $\mathcal{A}$ be a separable Γ - C^* -algebra, (Y, η) a measurable Γ -space, and $\mathcal{C}_{Y, \mathcal{A}}$ the associated collection of measurable maps. Given $\mu \in \text{Prob}(\Gamma)$, there exists $\varphi \in \mathcal{C}_{Y, \mathcal{A}}$ which is μ -stationary in the sense that*

$$(\mu * \varphi)(y)(a) := \sum_{s \in \Gamma} \mu(s) \varphi(s^{-1}y)(s^{-1}a) = \varphi(y)(a), \quad y \in Y_0, \quad a \in \mathcal{A}.$$

Proof. The set $\mathcal{C}_{Y, \mathcal{A}}$ is a convex set with respect to the pointwise addition induced from $S(\mathcal{A})$. In particular for $\varphi, \psi \in \mathcal{C}_{Y, \mathcal{A}}$ and $t \in (0, 1)$,

$$(t\varphi + (1-t)\psi)(y)(a) = t\varphi(y)(a) + (1-t)\psi(y)(a), \quad y \in Y_0, \quad a \in \mathcal{A}.$$

Since $\mathcal{A}$ is separable, there is an affine bijection between $\mathcal{C}_{Y, \mathcal{A}}$ and the convex set of unital completely positive maps $E : \mathcal{A} \rightarrow L^\infty(Y, \eta)$ (see [3, Theorem 2.10]). The collection of all unital completely positive maps $E : \mathcal{A} \rightarrow L^\infty(Y, \eta)$ is compact when endowed with the point ultraweak topology (see [5, Theorem 1.3.7]). This induces on $\mathcal{C}_{Y, \mathcal{A}}$ a topology which makes it compact. To sum up, $\mathcal{C}_{Y, \mathcal{A}}$ is a compact convex Γ -space. Given $\mu \in \text{Prob}(\Gamma)$, consider the induced affine continuous map $M_\mu : \mathcal{C}_{Y, \mathcal{A}} \rightarrow \mathcal{C}_{Y, \mathcal{A}}$ defined by $\varphi \mapsto \mu * \varphi$. The claim now follows from the Markov-Kakutani fixed point theorem. $\square$

Now, given a μ -stationary element in $\mathcal{C}_{Y, \mathcal{A}}$, using the affine bijection between $\mathcal{C}_{Y, \mathcal{A}}$ and unital completely positive maps $E : \mathcal{A} \rightarrow L^\infty(Y, \eta)$, we obtain a μ -stationary u.c.p map $E : \mathcal{A} \rightarrow L^\infty(Y, \eta)$. In subsequent work, we will investigate further the existence and uniqueness of such μ -stationary maps.

Remark 3.13. It follows from Proposition 3.12 that we can find a μ -stationary η -a.e defined μ -stationary measurable map $\varphi : Y \rightarrow \text{Prob}(X)$. Moreover, if we let $\nu = \text{bar}(\phi\eta)$, it is easy to see that $\nu = \int_Y \varphi(y) d\eta(y)$. We do not know if φ is the disintegration map associated with some factor map $\pi : (X, \nu) \rightarrow (Y, \eta)$.

4. RELATIVE PROXIMALITY

Figure 1: Proposition 3.12 and Remark 3.13 in arXiv:2405.17122, PDF page 12.

Proof. Let $\Gamma = \mathbb{Z}/2\mathbb{Z}$ and $\mu = \frac{1}{2}(\delta_e + \delta_g)$. This measure is admissible and non-degenerate: it is absolutely continuous with respect to Haar measure and its support generates Γ . Give every space below the trivial Γ -action.

For the first example, let $X = \{*\}$ with its unique probability measure and let $Y = \{0, 1\}$ with $\eta = \frac{1}{2}(\delta_0 + \delta_1)$. There is only one map $\varphi : Y \rightarrow \text{Prob}(X)$, namely

$$\varphi(0) = \varphi(1) = \delta_*.$$

It is μ -stationary because both actions are trivial. Its barycenter is $\nu = \delta_*$. Any measurable map $\pi : X \rightarrow Y$ is constant, so $\pi_*\nu$ is a Dirac mass and cannot equal η . Thus there is no factor map, and in particular φ is not the disintegration of one.

For a version in which factors do exist, take $X = Y = \{0, 1\}$ and $\nu = \eta = \frac{1}{2}(\delta_0 + \delta_1)$. Define the constant field

$$\varphi(0) = \varphi(1) = \nu.$$

Again it is stationary, and its barycenter is ν . A map $\pi : (X, \nu) \rightarrow (Y, \eta)$ is measure preserving only if it is a bijection. Its two fibers are singletons, and uniqueness of finite disintegration gives

$$\nu_y = \delta_{\pi^{-1}(y)}.$$

But $\varphi(y) = \nu$ gives each singleton mass $1/2$, so it is not this disintegration. This proves both assertions. $\square$

4 Exact obstruction and scope

The second example can be checked without invoking uniqueness: a conditional measure over y must assign mass one to $\pi^{-1}(y)$, while the constant uniform field assigns that singleton mass $1/2$. Thus the missing ingredient is exactly fiber support (or, equivalently in this finite setting, the idempotent/conditional-expectation structure of the Markov kernel).

The singleton example refutes the existence claim using only the hypotheses of Proposition 3.12. The equal-cardinality example additionally shows that, even when an ambient factor is available, an arbitrary stationary fixed point need not be its disintegration. The result does not rule out positive theorems under added compatibility or fiber-support hypotheses, nor does it classify when one can select a different stationary fixed point that is a disintegration.

5 Verification and novelty notes

All stationarity equations reduce to identities because the actions are trivial. All measures and pushforwards are displayed explicitly, and the disintegration calculation is finite. No computational experiment is used.

A bounded official-arXiv search through 11 August 2026 used arXiv id 2405.17122, the exact title, and combinations of “stationary measurable map”, “disintegration”, and “factor map”. It found the source paper but no later answer to Remark 3.13. Expert review should verify that the literal scope of the remark is the hypotheses of Proposition 3.12; the final scope paragraph separates this from possible strengthened formulations.

References

- [1] T. Amrutam, M. Klötzer, and H. Oppelmayer, *Relative stationary dynamical systems*, arXiv:2405.17122.